AI-Enabled Citizen Connect and Grievance Redressal System for Smart and Transparent Rural Governance

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Abstract*— Rural governance in developing regions faces persistent challenges such as delayed grievance redressal, lack of transparency, limited citizen participation, and inefficient communication between local authorities and residents. Although several e-governance initiatives exist, many systems remain fragmented, document-heavy, and inaccessible to digitally less-skilled citizens. To address these limitations, this paper presents GramLink-AI, an intelligent rural governance and citizen connect system powered by Artificial Intelligence. The proposed system applies Natural Language Processing (NLP) techniques to analyze citizen complaints and service requests submitted through a digital platform. A Bidirectional Long Short-Term Memory (BiLSTM) deep learning model is employed to classify grievances, identify user intent, and automatically route requests to the appropriate administrative department. By learning patterns from historical complaint data, the BiLSTM model improves accuracy in issue categorization, urgency detection, and response handling. GramLink-AI simplifies the grievance submission process, reduces manual workload for officials, and enables faster and more structured service delivery. The system enhances transparency, encourages citizen participation, and supports data-driven decision-making in rural administration. Overall, GramLink-AI demonstrates how AI and deep learning can be effectively applied to strengthen smart village initiatives and improve the efficiency of rural governance systems. Experimental evaluation indicates improved grievance handling efficiency and reduced manual intervention compared to traditional approaches.

Keywords*— *Rural E-Governance, Grievance Management System, Natural Language Processing, BiLSTM, Smart Village

I. INTRODUCTION

In India, a large portion of the population resides in rural areas, where the Gram Panchayat plays a vital role in local administration and public service delivery. The Gram Panchayat is responsible for implementing government schemes, maintaining village records, addressing citizen complaints, and supporting overall rural development. However, many rural regions still rely on manual and paper-based processes, which often result in slow grievance redressal, limited transparency, and inefficient governance.

Villagers frequently face challenges such as lack of awareness about government schemes, difficulty in tracking village development activities, and absence of an easy mechanism to submit and follow up on complaints. These limitations weaken communication between citizens and Panchayat authorities and reduce public participation in local governance. Although some digital solutions exist, they are often complex and not well suited for users with limited technical knowledge.

To address these challenges, this paper proposes GramLink-AI, an intelligent digital platform that directly connects citizens with the Gram Panchayat. The system enables users to view government schemes, monitor local development projects, submit complaints, and receive updates related to village activities and employment opportunities. Role-based access is provided for citizens, Panchayat members, and administrators to ensure secure and organized information management.

GramLink-AI is designed to be simple, user-friendly, and multilingual, making it accessible to a wide range of rural users. By automating grievance analysis and streamlining communication, the system improves transparency, reduces administrative delays, and encourages active citizen participation in village decision-making. The key contributions of this paper include: (i) an AI-enabled grievance redressal system using a BiLSTM model for intent-based complaint classification, (ii) intelligent prioritization and department-level routing of grievances using NLP techniques, and (iii) a centralized digital platform that enhances transparency, citizen engagement, and administrative efficiency in Gram Panchayat governance.

II. LITERATURE REVIEW

Rural governance in India faces several challenges, including limited access to government schemes, inefficient grievance redressal, and lack of timely communication between officials and citizens. Over the years, various digital initiatives have been introduced to address these issues, including e-Panchayat systems and state-level citizen service platforms. This literature survey reviews existing solutions, highlights their limitations, and identifies areas where a unified, citizen-focused platform like GramLink-AI can make a significant impact.

Recent developments in e-governance have played a significant role in improving rural administration and public service delivery in India. Deskar [1] discussed the impact of e-governance initiatives on rural development, emphasizing that digital platforms reduce administrative delays and improve villagers' access to government schemes. Similarly, Jonwal [2] highlighted the concept of smart villages and the role of ICT in enhancing service delivery, promoting informed decision-making, and supporting sustainable rural development.

Research on E-Gram Panchayat systems shows that digitization improves accountability and record management. Thorat et al. [3] focused on enhancing transparency in rural administration through digital Panchayat systems, demonstrating that online grievance handling and digital record maintenance significantly reduce corruption and improve trust in governance. Mane et al. [4] proposed an E-Gram Panchayat Management System that simplifies administrative processes such as complaint registration, scheme management, and citizen communication.

Grievance redressal remains a critical component of rural governance. Mishra et al. [5] presented an adaptive grievance redressal framework that improves response time and service efficiency by automating complaint classification and tracking. Salokhe et al. [6] discussed a Digital Gram Panchayat model that provides online access to complaints, schemes, and public notices, reducing the need for physical visits to Panchayat offices.

ICT-based governance solutions have been widely explored to increase citizen engagement. Sawane et al. [7] explained how E-Gram Panchayat systems improve accessibility and participation. Siddiqui et al. [8] proposed a Digital Gram Panchayat platform that supports online services and improves coordination between officials and villagers. Dhamale and Patil [9] designed an intelligent system emphasizing user-friendly interfaces and structured data management, while Dhage et al. [10] proposed an automated management system to reduce manual workload in Panchayat operations.

Other researchers have highlighted innovative approaches in digital governance. Jindal and Ojha [11] described the e-Panchayat initiative as a transformative model for rural administration, fostering transparency and efficiency. Bhagat and Niranjane [12] explored smart e-governance solutions for Grampanchayats, emphasizing digital service delivery and citizen-centric approaches. Maitra et al. [13] introduced a smart village system that integrates ICT tools to support rural development and improve resource management. Vora et al. [14] investigated the role of smart machines and applications in rural governance, demonstrating potential improvements in service efficiency and citizen engagement.

Dhage, Wathore, and Jagtap [15] further developed E-Gram Panchayat management systems, focusing on operational automation and enhanced administrative efficiency. Sangole and Bhura [16] emphasized the importance of sustainable development through e-governance, advocating for platforms that strengthen citizen participation, ensure accountability, and improve access to government schemes.

Overall, the reviewed literature highlights that existing digital governance initiatives enhance transparency, accountability, and citizen participation. However, many systems still lack intelligent complaint classification, multilingual support, and advanced automation. These gaps provide an opportunity for GramLink-AI to integrate AI and NLP techniques, delivering an intelligent, accessible, and citizen-centric platform that improves grievance handling, information dissemination, and administrative efficiency in rural governance.

III. PROPOSED METHODOLOGY

GramLink-AI is designed as an intelligent web and mobile-based governance platform aimed at improving public engagement, transparency, and grievance redressal in rural areas. The system enables villagers to conveniently access information about government schemes, monitor local development activities, submit grievances, and participate in community-related events through a simple and intuitive interface. At the same time, Panchayat officials are provided with a secure and centralized platform to manage complaints, update records, publish announcements, and analyze governance-related data. To ensure inclusivity across diverse rural populations, the system supports multilingual interaction and user-friendly design.

The operational workflow of GramLink-AI begins when a citizen submits a complaint or service request in the form of textual input through the web or mobile application. The system is designed to simplify the grievance submission process so that citizens can clearly describe local issues such as road damage, water supply problems, sanitation concerns, or streetlight failures. This structured text-based input ensures uniform data processing and improves the reliability of complaint analysis.

Once a complaint is submitted, the input text undergoes preprocessing to enhance data quality. This preprocessing includes removal of unnecessary symbols, normalization of text, and conversion into a standardized format suitable for machine learning analysis. These steps ensure consistency across different user inputs and improve the accuracy of the intelligent processing stage.

The processed text is then analyzed using a Bidirectional Long Short-Term Memory (BiLSTM) deep learning model integrated with an attention mechanism. The BiLSTM architecture enables the system to capture contextual dependencies from both preceding and succeeding words, allowing a deeper understanding of the intent and meaning of citizen complaints. The attention mechanism highlights

important keywords related to the type of issue, urgency indicators, and emotional tone, further improving classification performance.

Based on this intelligent analysis, the system automatically predicts the appropriate government department responsible for addressing the complaint, determines the urgency level of the issue, and identifies the sentiment of the citizen. All complaint details and prediction results are stored in a centralized database, enabling real-time tracking, feedback analysis, and performance monitoring. The complete workflow of the proposed system is illustrated in Fig. 1, demonstrating the interaction between citizens, AI components, and Panchayat authorities. This methodology ensures efficient, transparent, and citizen-centric grievance handling, supporting initiatives such as Digital India and Smart Village development.

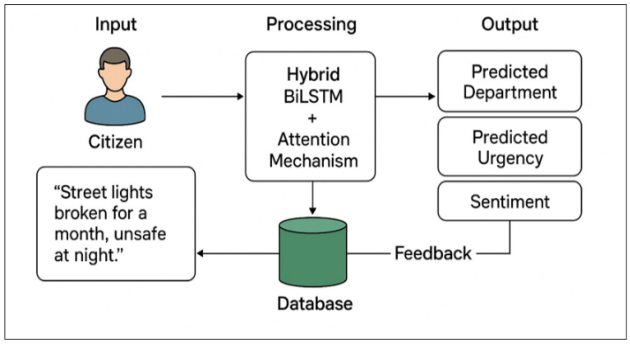

Fig. 1. Proposed System Architecture

IV. Implementation

The implementation of GramLink-AI – Intelligent Rural Governance and Citizen Connect System was carried out using a modular and scalable architecture to ensure ease of use, efficient processing, secure data management, and future extensibility. The system integrates modern web technologies, backend services, database management systems, and artificial intelligence models to deliver a comprehensive digital governance solution tailored specifically to rural administrative requirements. The overall design emphasizes simplicity, accessibility, and transparency, making it suitable for users with varying levels of digital literacy.

The user interface of GramLink-AI was developed using HTML, CSS, Bootstrap, and JavaScript, providing a responsive and intuitive platform accessible through both web and mobile browsers. Bootstrap was utilized to ensure compatibility across different screen sizes and devices, enabling consistent user experience on smartphones, tablets, and desktop systems. JavaScript was used to implement interactive functionalities such as real-time form validation, dynamic complaint tracking, and notification alerts. Citizens can register on the platform, submit grievances, and track the real-time status of their complaints, while administrators and Panchayat officials access a dedicated dashboard for grievance monitoring, response management, and system oversight.

The backend of the system was implemented using Java-based server technologies and RESTful APIs, which manage core business logic, user authentication, request handling, and secure communication between the frontend and the AI processing module. REST APIs facilitate modular integration and seamless data exchange between system components, improving maintainability and scalability. Secure authentication mechanisms and role-based access control ensure that only authorized users can access sensitive administrative data and perform privileged operations.

For data storage and management, a hybrid approach using SQL databases and Firebase was adopted. The SQL database stores structured information such as user profiles, grievance details, department assignments, urgency levels, and resolution status, ensuring data integrity and efficient querying. Firebase supports real-time data synchronization, authentication services, and notification delivery, allowing citizens and officials to receive instant updates regarding grievance progress, status changes, and administrative responses.

The intelligent core of GramLink-AI is powered by a Bidirectional Long Short-Term Memory (BiLSTM) deep learning model with an attention mechanism, implemented using Python. Before analysis, submitted grievance text undergoes preprocessing and is converted into numerical representations using word embedding techniques. The BiLSTM model captures contextual information from both past and future word sequences, enabling accurate understanding of complaint intent, while the attention mechanism emphasizes key terms related to issue type and urgency. The model classifies grievances, predicts urgency levels, and identifies sentiment, thereby automating grievance routing and prioritization. This intelligent processing significantly reduces manual workload, improves response time, and enhances decision-making for administrative authorities.

All predictions and grievance-related data are stored in a centralized database, enabling real-time monitoring, performance analysis, and feedback-driven system improvements. The system was tested using realistic rural governance scenarios to evaluate classification accuracy, response relevance, and overall system reliability. GramLink-AI was deployed locally during development and is designed to be cloud-ready, allowing future deployment on platforms such as AWS or Google Cloud. Its modular architecture supports easy expansion for multilingual support, additional governance services, and integration with existing e-governance infrastructures.

The GramLink-AI system also provides a centralized administrative dashboard that enables Panchayat officials to efficiently monitor and manage public grievances. The dashboard displays key statistics such as total grievance submissions, resolved cases, pending complaints, and escalated issues, along with recent complaint details and status updates. This visual interface supports quick decision-making, improves transparency, and enhances the overall efficiency of grievance handling. The administrative dashboard interface is shown in Fig. 2.

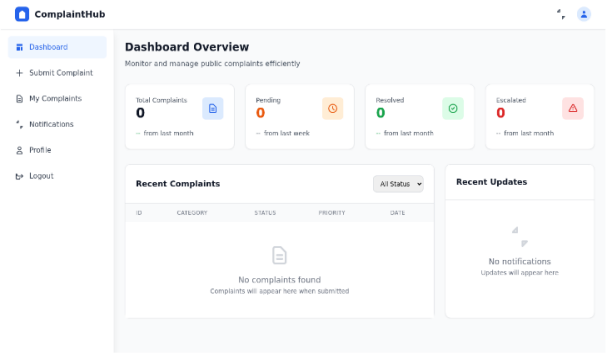

Fig. 2. Administrative Dashboard of GramLink-AI

V. Results and Discussion

The development of GramLink-AI resulted in a practical and intelligent digital governance platform that significantly improves rural administration by simplifying grievance submission and strengthening communication between citizens and Gram Panchayat authorities. The system effectively addresses major challenges such as delayed grievance redressal, lack of transparency, and limited citizen participation that are commonly observed in traditional manual governance processes. By integrating Natural Language Processing (NLP) and deep learning techniques, GramLink-AI enables citizens to submit complaints easily through text-based inputs, making the platform accessible even to digitally less-skilled rural users. This approach ensures uniform data processing and contributes to faster grievance routing and improved responsiveness compared to conventional manual handling methods.

The BiLSTM-based grievance classification model demonstrated a strong ability to understand the contextual meaning and intent of citizen complaints. This enabled automatic identification of the relevant administrative department and estimation of grievance urgency, thereby reducing the need for manual intervention by officials. As a result, the overall workload of Panchayat staff was significantly reduced, while the speed and structure of grievance handling were improved. The use of centralized data storage further enhanced system transparency by allowing real-time tracking of complaint status and supporting data-driven decision-making through analysis of grievance trends and patterns.

During evaluation under realistic rural governance scenarios, the system exhibited reliable performance in complaint classification and routing. However, certain limitations were observed. Variations in multilingual inputs may affect classification accuracy, and system scalability requires further optimization to efficiently support a larger number of users and complaints. Despite these challenges, the obtained results indicate that GramLink-AI is an effective and scalable solution for intelligent rural governance. The platform demonstrates strong potential to support Smart Village and Digital India initiatives by improving administrative efficiency, transparency, and citizen engagement in Gram Panchayat operations. The proposed system demonstrated improved grievance routing efficiency and reduced response delays when compared with conventional manual processes.

Table I. Functional Evaluation of GramLink-AI

Feature	Manual System	GramLink-AI
Complaint Submission	Physical visit	Online Text-based
Processing Time	High	Reduced
Department Mapping	Manual	Automatic
Tracking Status	Not Available	Real-time
Transparency	Low	High

VI. Conclusion

GramLink AI provides an efficient and intelligent solution for enhancing rural governance and citizen engagement. By integrating a centralized digital platform, the system addresses the limitations of traditional manual processes, enabling citizens to access government schemes, track projects, submit grievances, and obtain essential documents with ease. The role-based access and secure data management ensure transparency and privacy, while AI-driven recommendations and analytics support informed decision-making for officials. Additionally, features like community event updates and multilingual access promote inclusivity and active participation. The proposed system promotes inclusive digital governance by ensuring equitable access to public services for rural citizens. Overall, GramLink AI improves efficiency, accountability, and accessibility in Gram Panchayat operations, bridging the gap between citizens and local governance.

As part of future work, the system can be enhanced by strengthening multilingual NLP support and optimizing scalability using cloud-based deployment. These improvements will help GramLink-AI support larger user bases and further enhance its effectiveness in real-world rural governance scenarios.

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